Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER 2024

Name of the Course: B. Tech. (All Branches) Semester: I

Name of the Paper: Engineering Mathematics-I Paper Code: TMA-101

Time: 3 Hours Maximum Marks: 1(

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	
a)	Find the Rank of the given matrix: $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & -2 & 0 & -2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$	CO 1
b)	Find the values of the parameters λ and μ for which the system of equation $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ has (i) unique solution (ii) no solution (iii) infinite solution.	CO 1
c)	Examine whether the given matrix A is diagonalizable or not. If yes, find the modal/similarity matrix P and hence find D such that $D = P^{-1}AP$. $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$	CO 1
Q2	(20 marks)	
a)	If $f(x) = \sin^{-1} x$, $0 < a < b < 1$, use mean value theorem to prove that $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}$.	CO 2

b)	Evaluate the following limits: $lim_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{sin^2 x}\right)$.	
c)	The function $f(x) = sin\, x$ is approximated by Taylor's polynomial of degree three about the point $x = 0$. Find 'c' such that the error satisfies $\lvert R_3(x)\rvert \leq 0.001$ for all x in the interval $[0, c]$.	
Q3	(20 marks)	
a)	If $y = e^{m cos^{-1} x}$ then calculate $y_n(0)$.	CO 3
b)	If $u = sin^{-1}\left(\frac{x^{1/4}+y^{1/4}}{x^{1/6}+y^{1/6}}\right)$, prove that : $$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{1}{144} tanu\,(tan^2 u - 1)$$	
c)	The temperature T at any point (x, y, z) in space is $T(x, y, z) = kxyz^2$ where k is constant.Find the highest temperature on the surface of the sphere $x^2 + y^2 + z^2 = a^2$	
Q4	(20 marks)	
a)	Prove that : $$\int_0^{\pi/2} sin^m\theta cos^n\theta\, d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}$$ Hence evaluate $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.	CO 4/CO 5
b)	Change the order of integration $\int_0^2 \int_{\sqrt{4-x^2}}^{4-x} f(x, y)dy\, dx$.	
c)	Evaluate the triple integral of the function $f(x, y, z) = x^2$ over the region V enclosed by the planes $x = 0,\ y = 0,\ z = 0$ and $x + y + z = a$.	
Q5	(20 marks)	
a)	Show that $div\ (grad\ r^n) = n\,(n + 1)r^{n-2}$ Hence show that $\nabla^2\left(\frac{1}{r}\right) = 0$	CO 5/CO 6
b)	Apply Green's theorem to evaluate: $\oint_C [(2x^2 - y^2)dx + (x^2 + y^2)dy]$, where C is the boundary of the area enclosed by the $x -$ axis and the upper half of circle $x^2 + y^2 = a^2$.	
c)	Apply Stokes' theorem to evaluate $\oint_C \vec{F}.d\vec{r}$, where $\vec{F} = y^2\hat{\imath} + x^2\hat{\jmath} - (x + z)\hat{k}$ and C is the boundary of the triangle with vertices at (0,0,0), (1,0,0), (1,1,0).	